

Конспект «Расширенные возможности SQL»

1. INNER JOIN

```
SELECT f.title, CONCAT(a.last_name, ' ', a.first_name) AS  
actor_name  
FROM film f  
INNER JOIN film_actor fa ON fa.film_id = f.film_id  
INNER JOIN actor a ON a.actor_id = fa.actor_id;
```

2. LEFT JOIN

```
SELECT CONCAT(c.last_name, ' ', c.first_name), c2.city  
FROM customer c  
LEFT JOIN address a ON a.address_id = c.address_id  
LEFT JOIN city c2 ON c2.city_id = a.city_id;
```

3. RIGHT JOIN

```
SELECT CONCAT(c.last_name, ' ', c.first_name), c2.city  
FROM customer c  
RIGHT JOIN address a ON a.address_id = c.address_id  
RIGHT JOIN city c2 ON c2.city_id = a.city_id;
```

4. FULL JOIN

```
SELECT r.rental_id, r.rental_date, p.payment_id, p.payment_date,  
p.amount  
FROM rental r  
FULL JOIN payment p ON p.rental_id = r.rental_id;
```

5. CROSS JOIN

```
SELECT c.city, c2.city
FROM city c
CROSS JOIN city c2
WHERE c.city > c2.city;
```

```
SELECT c.city, c2.city
FROM city c, city c2
WHERE c.city > c2.city;
```

6. UNION

```
SELECT color_1 FROM table_1
UNION
SELECT color_2 FROM table_2;
```

7. UNION ALL

```
SELECT color_1 FROM table_1
UNION ALL
SELECT color_2 FROM table_2;
```

8. EXCEPT

```
SELECT color_1 FROM table_1
EXCEPT
SELECT color_2 FROM table_2;
```

9. Агрегатные функции

```
SELECT COUNT(1)
FROM film
WHERE LOWER(LEFT(title, 1)) = 'a';
```

10. GROUP BY

```
SELECT customer_id, MONTH(payment_date), COUNT(payment_id),
SUM(amount)
FROM payment
GROUP BY customer_id, MONTH(payment_date);
```

11. HAVING

```
SELECT CONCAT(c.last_name, ' ', c.first_name), COUNT(r.rental_id)
FROM rental r
JOIN customer c ON r.customer_id = c.customer_id
GROUP BY c.customer_id
HAVING COUNT(r.rental_id) > 40;
```

12. Подзапросы

```
SELECT MONTH(payment_date),
COUNT(payment_id) / (SELECT COUNT(1) FROM payment) * 100
FROM payment
GROUP BY MONTH(payment_date);
```

13. CASE

```
SELECT customer_id, SUM(amount),
CASE
    WHEN SUM(amount) > 200 THEN 'Good user'
    WHEN SUM(amount) < 200 THEN 'Bad user'
    ELSE 'Average user'
END AS good_or_bad
FROM payment
GROUP BY customer_id
ORDER BY SUM(amount) DESC
LIMIT 5;
```

14. IFNULL

```
SELECT CONCAT(c.last_name, ' ', c.first_name) AS user,
IFNULL(SUM(p.amount), 0)
FROM customer c
LEFT JOIN (
    SELECT *
    FROM payment
    WHERE DATE(payment_date) = '2005-06-18') p
ON p.customer_id = c.customer_id
GROUP BY c.customer_id
```

15. COALESCE

```
SELECT rental_id,  
COALESCE(DATEDIFF(return_date, rental_date), DATEDIFF(NOW(),  
return_date),  
DATEDIFF(NOW(), rental_date)) AS diff  
FROM rental
```